

Hip flexion strength remains decreased in anterior cruciate ligament reconstructed patients at one-year follow up compared to healthy controls

Georgios Mouzopoulos · Rainer Siebold ·
Mathaios Tzurbakis

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Abstract

Purpose The aim of this study was to evaluate hip muscle flexion strength in patients who underwent anterior cruciate ligament (ACL) reconstruction with patellar or hamstrings tendon graft and compare them with healthy controls.

Methods Sixty-eight male ACL deficient patients who underwent reconstruction (36 patients with hamstrings tendon graft and 32 patients with patellar tendon graft; randomized selection of type of graft) in our department, between July 2011 and July 2012, were enrolled in this randomised prospective study. Also 64 healthy male weekend athletes participated in the study as a control group. Hip flexor strength was measured by isokinetic dynamometer (Biodex), pre-operatively and one year after reconstruction. The statistical packet STATA 8.0 was used and statistical significance was set at $p < 0.05$.

Results Hip flexion strength in ACL reconstructed patients either with patellar tendon or hamstrings grafts, one year after reconstruction is significantly decreased compared to healthy controls ($p < 0.0001$). Patients reconstructed with patellar tendon have stronger hip flexors than those reconstructed with hamstrings graft ($p < 0.0001$).

Conclusion Hip flexion strength remains decreased one year after ACL surgery compared to healthy controls. Patellar tendon could be the graft of choice for ACL reconstruction in athletes who must preserve stronger hip flexors.

Keywords Hip · Flexors · ACL · Reconstruction · Muscle · Strength

Introduction

Many studies have evaluated anterior cruciate ligament (ACL)-injured patients at various periods after reconstruction surgery and rehabilitation [1–5]. Most of these reports studied isolated knee joint function using isokinetic devices for muscle strength measurements. In contrast only few studies have provided evaluation of the biomechanics of the hip after ACL reconstruction [6–8]. Considering that hip functions influence various knee pathologies and it has been proposed that a proximal link could lead to a distal problem, it is worth evaluating hip muscle strength after ACL reconstruction [9].

Decreased hip flexor strength has been reported after ACL reconstruction in few studies [8, 10, 11]. However, to the best of our knowledge there is no study that reports hip flexor strength alterations in ACL deficient patients reconstructed with patellar tendon or hamstrings graft and compare them amongst themselves and to healthy subjects.

As a matter of fact the hip flexors' main function is to help the hip joint move properly in coordination with hip extensors, providing not only hip, but also knee joint stability [10]. Strong hip flexors resist against excessive hip extension which generates forward velocity during the first half of the stance, leading to increased loads applied on ACL [12]. So if hip flexor strength decreases after ACL reconstruction either with patellar or hamstrings graft, then specific strengthening programs should be recommended in order to avoid lower limb strength imbalances, reduce the risk of fall and protect the graft. Besides, patellar or hamstring grafts used for ACL reconstruction may have a negative effect on hip flexion strength and this could affect the athletic performance [3].

G. Mouzopoulos · M. Tzurbakis
Orthopaedic Department of Evangelismos Hospital, Athens, Greece

G. Mouzopoulos · R. Siebold
Center for Hip-Knee-Foot Surgery & Sports Traumatology, ATOS
Klinik, Heidelberg, Germany

G. Mouzopoulos (✉)
10 Pamassou St, 35100 Lamia, Greece
e-mail: gmouzopoulos@yahoo.gr

The primary purpose of this study was to evaluate hip flexion muscle strength in ACL reconstructed patients with patellar or hamstring tendon graft used and compare them with healthy controls. The hypothesis of the study was that hip flexion strength is decreased one year after ACL reconstruction either with patellar tendon or hamstrings graft compared to healthy controls. Also, reconstructed patients with patellar tendon have stronger hip flexors than those reconstructed with hamstrings graft.

Materials and methods

Patients and entry criteria

ACL deficient patients, who reconstructed between July 2011 and July 2012 in our department, were considered to be potential subjects for this prospective randomized study.

Inclusion criteria were ACL rupture of the dominant leg and willingness of patients to comply with the recommended six-month postoperative physiotherapy program. Additionally, the patients had to be 20–35 years of age with Tegner score >4 points.

Exclusion criteria were ACL rupture of the non-dominant leg, previous fracture or reconstructive procedure of lower extremity, concurrent injury of other knee ligament, patella baja, meniscus lesion and hip disorder.

Randomization between ACL reconstructions with patellar tendon (PT group) or hamstrings tendon (HS group) grafts was performed in the operating room, following the arthroscopic procedure. Block randomization was used. Potential patients were divided into m blocks of size $2n$, and then each block was randomized such that n patients were allocated to patellar tendon graft and n to hamstrings tendon graft.

Example:

Assume the following types of treatment: P=ACL reconstruction with patellar tendon graft, H=ACL reconstruction hamstrings tendon graft and block size is $2 \times 2 = 4$.

Possible treatment allocations within each block were:

(1) PPHH, (2) HHPP, (3) PHPH, (4) HPHH, (5) PHHP, (6) HPPH

Seventy-six individuals were recruited to participate in this trial, but two failed to comply with physiotherapy recommendations, three denied to continue the measurements until the end and three could not be reached for follow up and were excluded. Finally, 68 ACL deficient male patients (age, 26.2 ± 5.6 years; weight, 78.2 ± 9.8 kg) met the inclusion criteria and were enrolled in the study. Thirty-two patients were reconstructed with

patellar tendon graft and 36 with hamstrings tendon graft. Also, 68 healthy controls (age, 25.7 ± 3.4 years; weight, 75.7 ± 3.9 kg) with no previous injury and randomly selected among 500 weekend athletes (one out of every seven listed patients was selected) participated in the study in order to provide comparison between ACL deficient and healthy subjects.

All participants provided informed consent before participating. The study was performed in accordance with the Declaration of Helsinki and was authorized by the local research and ethics committee.

Treatment

All ACL injured patients followed the same preoperative physiotherapy program for quadriceps strengthening, edema diminishing and achievement of complete range of knee joint motion, according to Pandy and Shelbourne recommendations [13].

ACL reconstructions were performed arthroscopically with patellar tendon or hamstrings graft. Middle third bone–patellar–bone and quadrupled hamstring graft fixed with endobutton were prepared as described elsewhere [14, 15]. The ACL reconstructions were performed on average 32 ± 4.1 days after ACL rupture in both PT and HS groups and by the same surgical team.

Postoperatively, the ACL reconstructed patients were recommended to follow the same physiotherapy program according to Gerber et al. [16]. The recommendations of this program are listed below:

First day–third week: weight bearing as tolerated after patellar tendon procedure and touch-down weight bearing after hamstring, range of motion exercises, straight leg strengthening, gait training

Fourth week–12th week: endurance training (bicycling), progressive resistance training (leg presses, calf presses, mini-squats, hamstring curls), balance exercises

13th week–24th week: progressive resistance and endurance training, jogging / running progression, advanced strengthening (hip–knee–ankle)

Outcome measures

Isokinetic hip muscle flexion strength was performed. ACL deficient patients reported for testing on two occasions, just one day before surgery and one year postoperatively at the end of the study. Healthy controls reported for testing on a single occasion only at the end of the study.

Table 1 Preoperative and postoperative values of hip flexion strength of the dominant limb in the patellar tendon (PT) graft group at different angular velocities

Angular velocity measurement	Preoperative	Postoperative	p-value
120° eccentric	0.92±0.07	1.31±0.05	0.00001
60° eccentric	0.87±0.04	1.23±0.09	0.00001
120° concentric	0.81±0.05	1.2±0.05	0.00001
60° concentric	0.77±0.07	1.36±0.09	0.00001

Isokinetic muscle strength

Hip flexion strength was measured using the Biodex 6000 dynamometer (Biodex Medical System Inc., Shirley, New York) as it has been described by Perrin [17]. Three repetitions of hip flexor contraction at an angular velocity of 120/seconds and 60/seconds in a concentric and eccentric mode were performed. Flexion strength was measured in standing position. Flexion contractions were performed at neutral hip position.

The average peak torque of the three contractions of each participant was recorded and the data were normalized to body mass. Subjects were encouraged to stop the procedure immediately in case of pain or cramping. Two specialized physiotherapists performed the measurements. They also supervised the physiotherapy program postoperatively. At one-year follow up the reconstructed patients were measured first, and the controls last. The tests were carried out on both hips. Testing order (dominant or non dominant leg) was randomized prior to subject enrolment. All subjects were allowed a two minutes rest period between exercises.

Statistical analysis

SPSS version 15.0 (Chicago, IL) was used to provide statistical analysis. Descriptive statistics were run on all variables including mean and standard deviation. A paired sample t-test was used to detect significant differences between preoperative versus postoperative values in each group. A two-way ANOVA analysis was used to determine whether there were significant differences in terms of mean peak torque at any angle among the three groups (PT, HS and healthy controls),

Table 2 Preoperative and postoperative values of hip flexion strength of the dominant limb in the hamstrings tendon (HS) graft group at different angular velocities

Angular velocity measurement	Preoperative	Postoperative	p-value
120° eccentric	0.94±0.05	1.13±0.16	0.00001
60° eccentric	0.75±0.1	1.16±0.1	0.00001
120° concentric	0.78±0.09	1.05±0.22	0.00001
60° concentric	0.73±0.1	1.01±0.28	0.00001

either pre-operatively or postoperatively. Statistical significance was set at $p < 0.05$.

Results

Pre-operatively no significant difference in terms of hip flexion strength was found between PT and HS groups, regarding neither the dominant nor the non-dominant leg ($p > 0.05$).

Preoperative versus postoperative hip flexor strength of the dominant or non-dominant leg showed a significant increase in both groups one year after ACL reconstruction (Tables 1, 2, 3, and 4).

Although the hip flexion strength values of the dominant leg in the PT and HS groups were increased one year after ACL reconstruction, they remained significantly shorter than healthy controls (Figs. 1, 2, 3, and 4). Besides hip flexion strength was found significantly higher in the PT than the HS group one year postoperatively ($p < 0.0001$).

Also, no significant difference between the PT or HS groups and the healthy control group, regarding the non-dominant leg, was found one year postoperatively ($p > 0.05$).

Discussion

The notable point of our study was that the hip flexors strength is decreased after ACL reconstruction either with patellar tendon or hamstrings graft and a subsequent rehabilitation program, compared to healthy controls. Besides, significant differences between patients reconstructed with patellar tendon and those reconstructed with hamstrings was detected. Patients reconstructed with patellar tendon graft experienced significantly greater hip flexor strength values than the other patients. Pre-operative hip flexors strength in both dominant and non-dominant legs were lower than postoperatively, probably due to the lack of exercise as it usually happens with injured patients before operation and to the subsequent physiotherapy program followed after surgery [18].

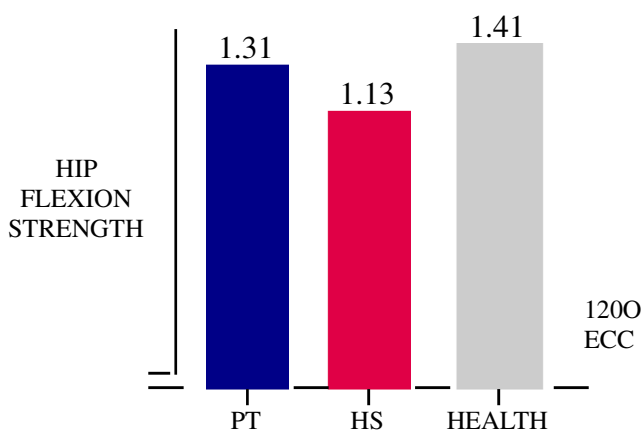
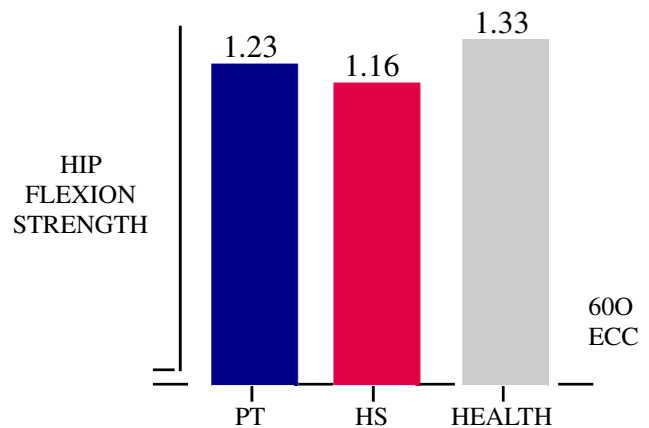
Table 3 Preoperative and postoperative values of hip flexion strength of the non dominant limb in the patellar tendon (PT) graft group at different angular velocities

Angular velocity measurement	Preoperative	Postoperative	p-value
120° eccentric	0.83±0.06	1.17±0.06	0.00001
60° eccentric	0.77±0.07	1.06±0.09	0.00001
120° concentric	0.79±0.07	0.98±0.11	0.00001
60° concentric	0.69±0.06	0.74±0.1	0.0088

Table 4 Pre-operative and postoperative values of hip flexion strength of the non dominant limb in the hamstrings tendon (HS) graft group at different angular velocities

Angular velocity measurement	Preoperative	Postoperative	p-value
120° eccentric	0.82±0.06	1.18±0.07	0.00001
60° eccentric	0.8±0.06	1.07±0.12	0.00001
120° concentric	0.77±0.09	0.96±0.12	0.00001
60° concentric	0.67±0.08	0.74±0.15	0.00001

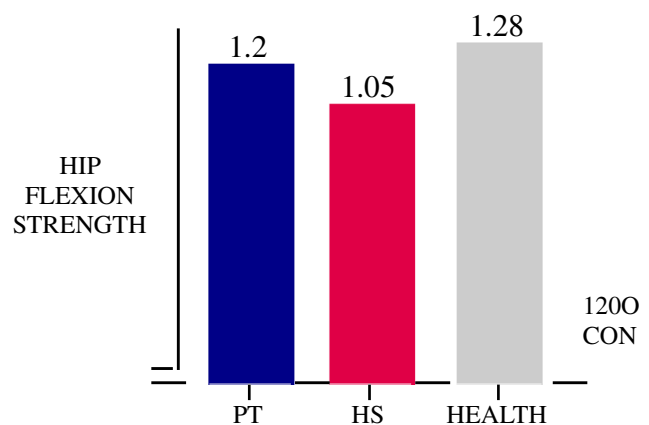
Karanikas et al. examined the hip muscles strength in 18 patients who underwent ACL reconstruction with patellar tendon and in 17 patients who were reconstructed with hamstrings [10]. The authors found significantly reduced maximal isokinetic moments at the hip flexors, as compared to the non-injured limb, at 12–24 months after surgery. In contrast to our results, the authors found no significant difference between patients reconstructed with patellar tendon and those reconstructed with hamstrings, during 12–24 months after ACL reconstruction. Probably differences in strength evaluation technique may contribute to different findings. First of all, Karanikas et al. reported the hip flexor strength values without normalizing these values to body mass. Second, they tested the hip flexion in supine position but we did this in standing position. Also in this last study the authors were looking for differences among injured and uninjured legs without specifying which leg was dominant or non-dominant. Besides, they did not mention if there was an equal distribution of dominant and non-dominant injured legs among the patients reconstructed with patellar tendon and those with hamstrings graft. However, it is well known that greater muscle strength in dominant than in non-dominant legs could exist [13]. This heterogeneity between the two groups (patellar tendon or hamstrings graft) could lead to no significant differences regarding the hip flexion strength. In our study all the ACL

**Fig. 1** Postoperative hip flexion strength in dominant leg after ACL reconstruction with patellar tendon (PT) or hamstrings (HS) graft, compared to healthy controls performed under 120° eccentric contraction**Fig. 2** Postoperative hip flexion strength in dominant leg after ACL reconstruction with patellar tendon (PT) or hamstrings (HS) graft, compared to healthy controls performed under 60° eccentric contraction

deficient patients had injured the dominant leg, so there was homogeneity between the two groups.

A possible explanation of decreased hip flexor values in ACL reconstructed patients compared to healthy controls could be due to the bi-articular nature of the rectus femoris and to the quadriceps weakness or voluntary decrease of quadriceps action, following ACL surgery [10, 11]. As a matter of fact, lower extremity muscle weakness, particularly in the quadriceps, is commonly reported following ACL rupture and reconstruction, either with patellar tendon or hamstrings grafts due to insufficient postoperative rehabilitation or bone (tunnel drilling, notch plasty) and soft tissue (muscle, periosteum) damage [19–22].

The consequence of hip flexors strength deficit is gait impairment and as a result a great risk of fall, under specific circumstances [23]. During normal gait, the ankle plantar flexors produce vertical support, forward progression and hip extension [10]. The critical role of hip flexors is to resist this excessive hip extension [24]. In case of hip flexors

**Fig. 3** Postoperative hip flexion strength in dominant leg after ACL reconstruction with patellar tendon (PT) or hamstrings (HS) graft, compared to healthy controls performed under 120° concentric contraction

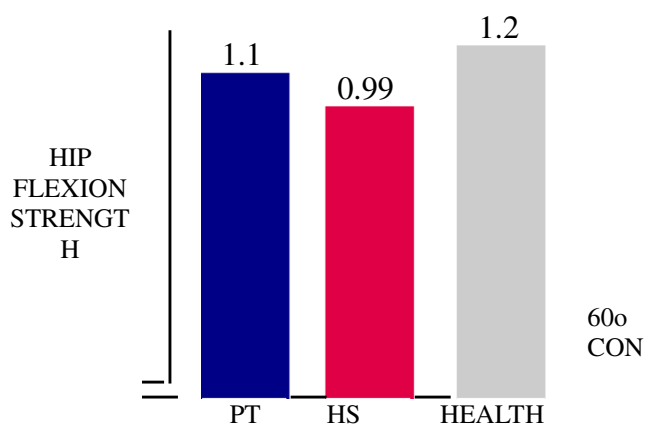


Fig. 4 Postoperative hip flexion strength in dominant leg after ACL reconstruction with patellar tendon (PT) or hamstrings (HS) graft, compared to healthy controls performed under 60° concentric contraction

weakness, a compensatory strategy takes place in order to decrease the demand on the hip flexors by increasing knee flexion angle, which decreases the excessive hip extension [23]. Also increased knee flexion angle is associated with decrease of ACL length, leading to less stress applied on the ligament [22, 25, 26].

It is worth mentioning here that physiotherapy programs should focus on hip flexors strengthening in order to avoid hip muscles strength imbalances, thus giving the chance to protect the graft. Hip agonist and antagonist muscle group co-activation plays an important role in stabilizing the knee joint [27].

Limitations of the study include that the subjects were amateur male athletes, so the results could not be generalized to professional or female athletes. Moreover, isokinetic assessment of muscle strength is a static method and it does not allow the measurement of the muscular forces applied in dynamic conditions of daily activities.

A further suggestion for study is to replicate the present trial including other hip muscles, also knee and ankle muscles strength measurements, in order to examine the whole leg biomechanics and kinematics as an entire unit after ACL reconstruction.

Conclusion

In summary, hip flexion strength of reconstructed ACL deficient patients, either with patellar tendon or hamstring grafts is significantly decreased, compared to healthy controls one year after surgery. Patients reconstructed with patellar tendon have stronger hip flexor strength than those reconstructed with hamstrings graft.

Conflict of interest None declared.

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